From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2013

FULL TEST – I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-B**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (13 – 18) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 marks** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 mark** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

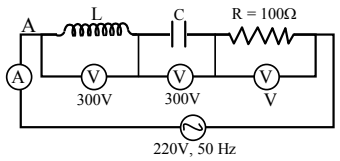
PART – I

SECTION – A

Single Correct Choice Type

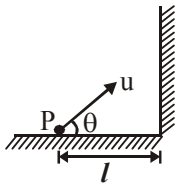
This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In the circuit shown below, what will be the reading of the voltmeter and ammeter?



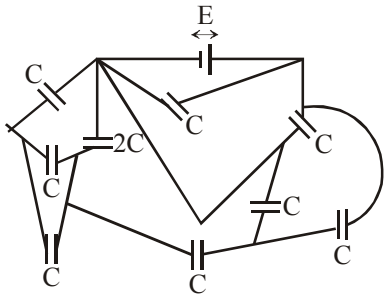
(A) 800 V, 2A (B) 300 V, 2A
(C) 220 V, 2.2A (D) 100 V, 2A

2. A ball of mass m is projected from a point P on the ground as shown in the figure. It hits a fixed vertical wall at a distance *l* from P. Choose the most appropriate option :



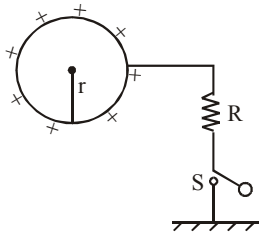
(A) the ball will return to the point P if *l* = half of the horizontal range.
(B) the ball will return to the point P if *l* ≤ half of the horizontal range.
(C) the ball can not return to the initial point if *l* > half of the horizontal range.
(D) the ball will return to the initial point, if the collision elastic and *l* < half of the range.

3 What is the equivalent capacitance across the battery?



(A) $\frac{79}{30}$C (B) $\frac{59}{30}$C
(C) $\frac{41}{30}$C (D) $\frac{21}{30}$C

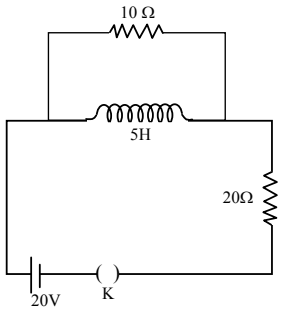
4. Initial charge on conducting sphere of radius r is Q_0. If S is closed at t = 0 then charge on the sphere at any time t is



(A) $Q_0 e^{-t/rR}$ (B) $Q_0 e^{\frac{-t}{4\pi\varepsilon_0 R}}$
(C) $Q_0 e^{-\frac{t}{4\pi\varepsilon_0 Rr}}$ (D) none of these

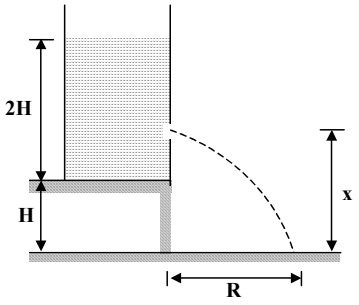
Rough work

5. Two resistances of 10 Ω and 20 Ω and an ideal inductor of inductance 5H are connected to a 20V battery through a key K, as shown in figure. The key is closed at t = 0. What is the final value of current in the 10Ω resistor?



(A) (2/3) A (B) (1/3) A
(C) (1/6) A (D) zero

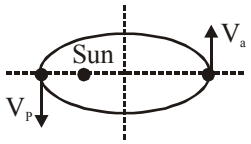
6. A tank is filled upto a height 2*H* with a liquid and is placed on a platform of height *H* from the ground. The distance *x* from the ground where a small hole is punched to get the maximum range *R* is



(A) H (B) 1.25 H
(C) 1.5 H (D) 2 H

7. The measure of radius of a sphere is (4.22 + 2%) cm. The volume of the sphere is
(A) $(315 \pm 6\%)\,\text{cm}^3$ (B) $(315 \pm 2\%)\ \text{cm}^3$
(C) $(315 \pm 4\%)\ \text{cm}^3$ (D) $(315 \pm 8\%)\ \text{cm}^3$

8. A planet moves around Sun in an elliptical orbit of eccentricity e. The ratio of the velocity at perigee V_p and at apogee V_a is given by



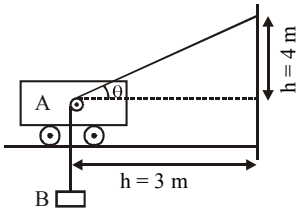
(A) $\frac{V_P}{V_a} = \frac{1+e}{1-e}$ (B) $\frac{V_P}{V_a} = \frac{1-e}{1+e}$
(C) $\frac{V_P}{V_a} = \sqrt{\frac{1+e}{1-e}}$ (D) $\frac{V_P}{V_a} = \sqrt{\frac{1-e}{1+e}}$

Rough work

Multiple Correct Answer(s) Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

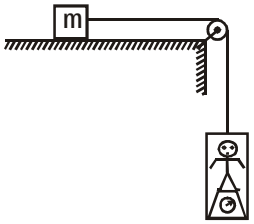
9. The string shown in the figure is passing over small smooth pulley rigidly attached to trolley *A*. If speed of trolley is constant and equal to V_A. Speed and magnitude of acceleration of block B at the instant shown in figure is



(A) $v_B = v_A$, $a_B = 0$ (B) $a_B = 0$

(C) $v_B = \frac{3}{5}v_A$ (D) $a_B = \frac{16v_A^2}{125}$

10. In the figure, a man of true mass *M* is standing on a weighing machine placed in a cabin. The cabin is joined by a string with a body of mass *m*. Assuming no friction, and negligible mass of cabin and weighing machine, the measured mass of man is (normal force between the man and the machine is proportional to the mass)



(A) measured mass of man is $\frac{Mm}{(M+m)}$ (B) acceleration of man is $\frac{mg}{(M+m)}$

(C) acceleration of man is $\frac{Mg}{(M+m)}$ (D) measured mass of man is M.

11. A charged particle of mass 2 kg and charge 2 C moves with a velocity $\vec{v} = 8\hat{i} + 6\hat{j}$ m/s in a magnetic field $\vec{B} = 2\hat{k}$ T. Then

(A) The path of particle may be $x^2 + y^2 = 25$. (B) The path of particle may be $x^2 + z^2 = 25$.

(C) The time period of particle will be 3.14 s. (D) None of these.

12. Choose the correct statement(s)

(A) The density of nuclear matter is independent of the size of the nucleus.

(B) The binding energy per nucleon, for nuclei of middle mass numbers, is about 8 MeV.

(C) A free neutron is unstable.

(D) A free proton is stable.

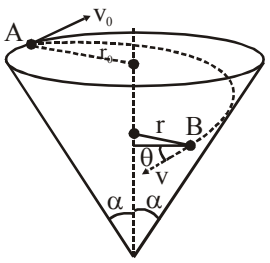
Rough work

Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 13 to 15

A small particle of mass m is given an initial velocity v_0 tangent to the horizontal rim of a smooth cone at a radius r_0 from the vertical centerline as shown at point A. As the particle slides to point B, a vertical distance h below A and a distance r from the vertical centerline, its velocity v makes an angle θ with the horizontal tangent to the cone through B.



13. The value of θ is

(A) $\cos^{-1}\dfrac{v_0 r_0}{\sqrt{v_0^2+2gh(r_0-h\tan\alpha)}}$ (B) $\cos^{-1}\dfrac{v_0 r_0}{\sqrt{v_0^2+2gh(r_0+h\tan\alpha)}}$

(C) $\cos^{-1}\dfrac{v_0 r_0}{\sqrt{v_0^2-2gh(r_0-h\tan\alpha)}}$ (D) $\cos^{-1}\dfrac{v_0 r_0}{r_0\sqrt{v_0^2+2gh}}$

14. The speed of particle at point B

(A) $\sqrt{v_0^2+2gh}$ (B) $\sqrt{v_0^2-2gh}$

(C) $\sqrt{v_0^2+gh}$ (D) $\sqrt{2v_0^2+2gh}$

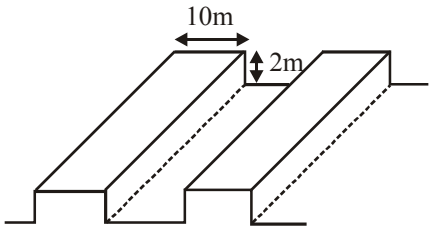
15. The minimum value of v_0 for which particle will be moving in a horizontal circle of radius r_0.

(A) $\sqrt{\dfrac{2gr_0}{\tan\alpha}}$ (B) $\sqrt{\dfrac{gr_0}{2\tan\alpha}}$

(C) $\sqrt{\dfrac{gr_0}{\tan\alpha}}$ (D) $\sqrt{\dfrac{4gr_0}{\tan\alpha}}$

Rough work

Paragraph for Question Nos. 16 to 18



The world is focusing its attention on renewable sources of energy like solar energy, wind energy, tidal energy and wave energy. These sources are non-polluting, do not cause the emission of greenhouse gases, or cause any large scale damage to the ecology or environment.

Waves on the surface of the ocean are a good source of power. To illustrate this, we calculate the mechanical energy carried by an average wave of crest 1m, wavelength 20 m and a period of 5 s. The wave profile is taken as approximately step-like, instead of a sinusoidal function.

A simple minded calculation gives us a contribution of 200 kW from the release of potential energy by such a wave over a 1 m wavefront.

16. The speed of the wave is
(A) 100 ms^{-1} (B) 4 ms^{-1}
(C) 0.25 ms^{-1} (D) none of these

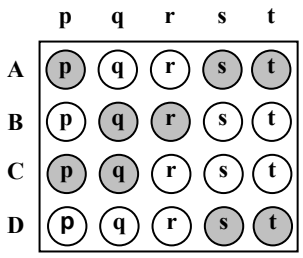
17. Wave energy provides an inexpensive source of power. In the paragraph above, only the potential energy carried by the wave was calculated. The contribution to power due to kinetic energy, assuming that all the water in the crest is moving forward at the speed of the wave gives us, over a 1 m wavefront approximately, of the order of
(A) 10 W (B) 100 W
(C) 10^3 W (D) 10^4 W

18. The momentum carried by the crest of the wave, per metre of the wavefront, is of the order of,
(A) 10^5 kg ms^{-1} (B) 10^2 kg ms^{-1}
(C) 10 kg ms^{-1} (D) 10^8 kg ms^{-1}

Rough work

SECTION - B
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. For the following statements, except gravity and contact force between the contact surfaces, no other force is acting on the body.

Column A		Column B	
(A)	When a sphere is in pure–rolling on a fixed horizontal surface.	(p)	Upward direction
(B)	When a cylinder is in pure rolling on a fixed inclined plane in upward direction then friction force acts in	(q)	$v_{cm} > R\,\omega$
(C)	When a cylinder is in pure rolling down a fixed incline plane, friction force acts is	(r)	$v_{cm} < R\omega$
(D)	When a sphere of radius R is rolling with slipping on a fixed horizontal surface, the relation between v_{cm} and ω is	(s)	No frictional force acts.

2. Match the column I with column II
In simple harmonic motion: x = 1.0 sin [12πt] and mass of particle executing SHM, m = 1/4 kg

Column A		Column B	
(A)	Frequency with which kinetic energy oscillates = …	(p)	$\frac{1}{12}$
(B)	Speed of particle is maximum at time = ….	(q)	$18\pi^2$
(C)	Maximum potential energy = …	(r)	12
(D)	Force constant k = …	(s)	$36\pi^2$

Rough work

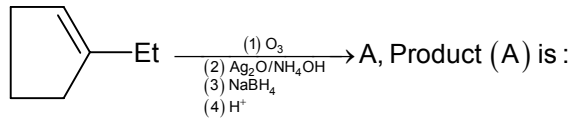
Chemistry

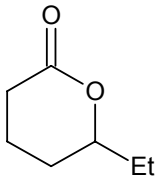
PART – II

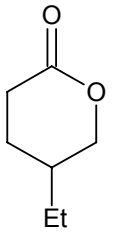
SECTION – A

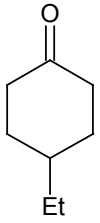
Straight Objective Type

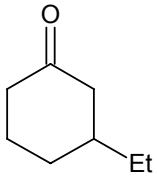
This section contains 8 multiple choice questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

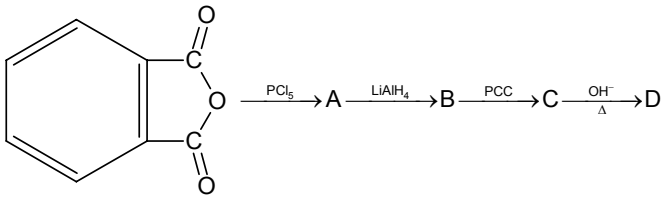
1. 

(A) 

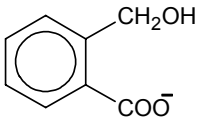
(B) 

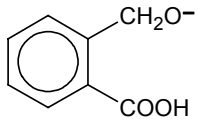
(C) 

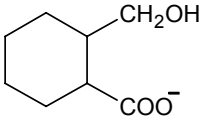
(D) 

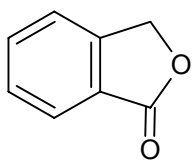
2. 

Product (D) is:

(A) 

(B) 

(C) 

(D) 

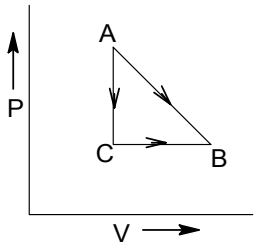
Rough Work

3. The two forms of D-glucopyranose obtain from the solution of D-glucose are called
(A) Isomer (B) Anomer
(C) Epimer (D) Enantiomer

4. The vapour pressure of pure benzene and toluene are 160 and 60 torr respectively. The mole fraction of toluene in vapour phase in contact with equimolar solution of benzene and toluene is:
(A) 0.50 (B) 0.6
(C) 0.27 (D) 0.73

5. Consider the modes of transformations of a gas from state 'A' to state 'B' as shown in the following P – V diagram. Which one of the following is true?

(A) $\Delta H = q$ along $A \rightarrow C$
(B) ΔS is same along both $A \rightarrow B$ and $A \rightarrow C \rightarrow B$
(C) W is same along both $A \rightarrow B$ and $A \rightarrow C \rightarrow B$
(D) W > O along both $A \rightarrow B$ and $A \rightarrow C$



6. Select the group of species in which all show trigonal bipyramidal geometry:
(A) PF_5, IF_5, XeF_4 (B) ClO_4^-, IF_7, CO_3^{2-}
(C) I_3^-, XeF_2, SF_4 (D) XeF_6, PF_6^-, ICl_2^+

7. Crystal field stabilization energy for high spin d^4 octahedral complex is:
(A) - 1.8 Δ_o (B) - 1. 6 Δ_o + P
(C) - 1.2 Δ_o (D) - 0.6 Δ_o

8. Solubility product of silver bromide is 5×10^{-13}. The quantity of potassium bromide (molar mass taken as 120 g mol^{-1}) to be added to 1 litre of 0.05 M solution of silver nitrate just to start the precipitation of AgBr is:
(A) 5×10^{-8} g (B) 1.2×10^{-10} g
(C) 1.2×10^{-9} g (D) 6.2×10^{-5} g

Rough Work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. Which of the following statement(s) is/are correct?
   (A) The coordination number of each type of ion in CsCl crystal is 8
   (B) A metal that crystallizes in bcc structure has a coordination number of 12
   (C) A unit cell of an ionic crystal shares some of its ions with other unit cells
   (D) The length of the unit cell in NaCl is 552 pm
   $\left(r_{Na^+} = 95\ pm, r_{Cl^-} = 181\ pm\right)$

10. Which of the following statements are correct for cis-1,2-dibromocyclopentane?
   (A) It contains two chiral centres, but is optically inactive
   (B) It can exist in two enatiomeric forms but cannot be optically active
   (C) It is a meso compound
   (D) It is with two chiral centres and is optically active

11. Which of the following pairs can be distinguished by using Lucas Test?
   (A) $PhCH_2OH$, CH_3CH_2OH (B) $PhCH_2OH$, PhOH
   (C) $(CH_3)_2CHOH$,$CH_3CH_2CH_2OH$ (D) $CH_3CH_2CH_2OH$, CH_3CH_2OH

12. Which reagent does not give oxygen as one of the products during oxidation with ozone?
   (A) SO_2 (B) $SnCl_2$ + HCl
   (C) H_2S (D) PbS

Rough Work

Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 13 to 15

For an electrode reaction written as

$M^{n+} + ne^- \longrightarrow M$

$E_{red} = E^o_{red} - \frac{RT}{nF}\ln\frac{1}{[M^{n+}]}$ (Nernst equation)

$= E^o_{red} - \frac{0.0591}{n}\log\frac{1}{[M^{n+}]}$ at 298 K

For the cell reaction, $aA + bB \rightleftharpoons xX + yY$

$E_{cell} = E^o_{cell} - \frac{RT}{nF}\ln\frac{[X]^x[Y]^y}{[A]^a[B]^b}$

$= E^o_{cell} - \frac{0.0591}{n}\log\frac{[X]^x[Y]^y}{[A]^a[B]^b}$ at 298 K

For pure solids, liquids or gases at 1 atm, molar concentration = 1

Standard free energy change $\Delta G^o = -nE^o_{cell}F$ where n is the number of electrons transferred in the redox reaction of the cell, E^o_{cell} is the standard emf of the cell. F stands for 1 Faraday, i.e. 96500 C mol^{-1} (approx.).

Standard free energy change $\Delta G^o = \frac{-2.303RT}{nF}\log K_{eq}$. Where K_{eq} is the equilibrium constant at TK. K_{eq} can be calculated from E^o_{cell} by using the relation, $E^o_{cell} = \frac{0.0591}{n}\log K_{eq}$

13. The e.m.f. of the cell
$Zn\,|\,Zn^{+2}(0.01\,M)\,||\,Fe^{+2}(0.001\,M)\,|\,Fe$
at 298 K is 0.2905 V. The value of the equilibrium constant for cell reaction is:
(A) $e^{\frac{0.32}{0.0295}}$ (B) $10^{\frac{0.32}{0.0295}}$
(C) $10^{\frac{0.26}{0.0295}}$ (D) $10^{\frac{0.32}{0.0591}}$

14. On the basis of information available from the reaction
$\frac{4}{3}Al + O_2 \longrightarrow \frac{2}{3}Al_2O_3,\ \Delta G = -827\ kJ\ mol^{-1}$
The minimum emf required to carry out an electrolysis of Al_2O_3 is (F = 96500 mol^{-1})
(A) 2.14 V (B) 4.28 V
(C) 6.42 V (D) 8.56 V

15. E^o for the cell, $Zn\,|\,Zn^{+2}_{(eq)}\,||\,Cu^{+2}_{eq}\,|\,Cu$; is 1.1 V at 25°C. The equilibrium constant for the cell reaction $Zn + Cu^{+2}_{(aq)} \rightleftharpoons Cu + Zn^{+2}_{(eq)}$ is of the order of
(A) 10^{-37} (B) 10^{37}
(C) 10^{-17} (D) 10^{17}

Rough Work

Paragraph for Question Nos. 16 to 18

In oxoacids of phosphorus, P is tetrahedrally surrounded by other atoms. All these acids contain one P = O and P – OH bond. The oxoacids in which phosphorus has lower oxidation state (less than +5) contain, in addition to P = O and P – OH bonds, either P – P (e.g., in $H_4P_2O_6$) or P – H (e.g., in H_3PO_2) bonds but not both. These acids in +3 oxidation state of phosphorus tend to disproportionate to higher and lower oxidation states. For example, orthophosphorous acid (or phosphorus acid) on heating disproportionate to give orthophosphoric acid (or phosphoric acid) and phosphine.

$4H_3PO_3 \longrightarrow 3H_3PO_4 + PH_3$

16. $H_3PO_2 + CuSO_4 \longrightarrow (X)$, a red ppt.

X is:

(A) Cu (B) Cu_2O

(C) CuO (D) Cu_2H_2

17. $P_4\,(\text{white}) + J(\text{an alkaline solution}) \longrightarrow K(\text{reducing gas}) + L$

L + dil. $H_2SO_4 \rightarrow$ N (ppt.) + M (oxyacids P)

N gives apple green colour in the flame.

Thus J, K, L, M and N respectively are:

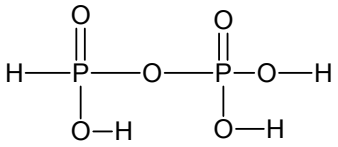
(A) $Ba(OH)_2$, PH_3, $Ba(H_2PO_2)_2$, H_3PO_2, $BaSO_4$

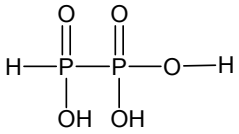
(B) $Ca(OH)_2$, P_2H_4, $Ba(H_2PO_2)_3$, H_3PO_2, $CaSO_4$

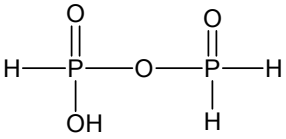
(C) $Ba(OH)_2$, PH_3, $Ba(H_2PO_2)_3$, H_3PO_3, $BaSO_4$

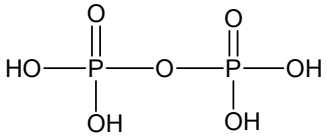
(D) $Ba(OH)_2$, P_2H_4, $Ba(H_2PO_2)_2$, H_3PO_2, $BaSO_4$

18. Which of the following represents the isopolyacid of phosphorus?

(A) 

(B) 

(C) 

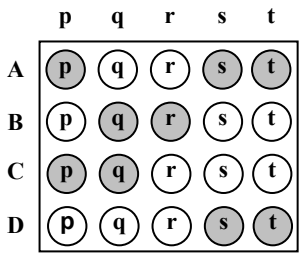
(D) 

Rough Work

SECTION-B
(Matrix Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



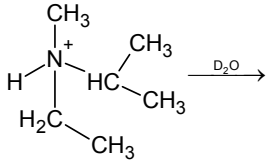
1. Acids given in Column – I are treated with OH^- and then with H^+, if required. The results are given in Column – II. Match correctly:

Column – I	Column – II
(A) 2-bormopropanoic acid	(p) Product is optically active
(B) 3-bromobutanoic acid	(q) Product shows geometrical isomerism
(C) 4-bromobutanoic acid	(r) Involves S_N2 attack
(D) 5-boromobutanoic acid	(s) Product contains a ring
	(t) Product contains – OH group

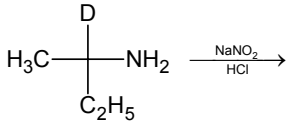
2. Match the reactions given in Column-I with the facts given in Column-II:

Column – I

(A)

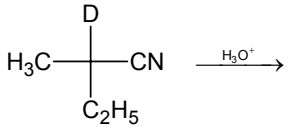


(B)



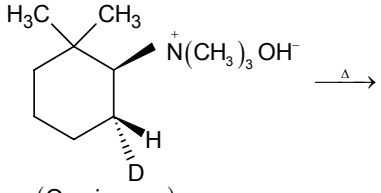
(One enantiomer)

(C)



(One enantiomer)

(D)



(One isomer)

Column – II

(p) Develops a racemic mixture

(q) An alkene is obtained

(r) Configuration is retained

(s) Product may contain deuterium

(t) No stereogenic centre in the product

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 8 multiple choice questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If a function y = f(x) is such that f′(x) < 0, then the number of integral values of 'a' for which the major axis of ellipse $f(a + 11)x^2 + f(a^2 + 2a + 5)y^2 = f(a + 11)f(a^2 + 2a + 5)$ becomes x–axis is
   (A) 1 (B) 2
   (C) 3 (D) 4

2. In a ΔABC, A, B, C are in AP and a, b, c are in GP then value of $a^3 + b^3 + c^3 – a^2b – b^2c – c^2a$ is
   (A) 0 (B) 1
   (C) 3 (D) 4

3. The four points A, B, C, D in space are such that angle ABC, BCD, CDA and DAB are all right angles, then
   (A) A, B, C, D cannot be coplanar (B) A, B, C, D are necessarily coplanar
   (C) A, B, C, D may or may not be coplanar (D) no such points A, B, C, D exist

4. b and c are arithmetic means between a and d (a > d > 0) and h and k are the geometric mean between a and d then
   (A) bc is always greater than hk (B) bc is always less than hk
   (C) bc may be equal to hk (D) none of these

5. If P be a point on ellipse $4x^2 + y^2 = 8$ with eccentric angle $\frac{\pi}{4}$. Tangent and normal at P intersects the axes at A, B, A′ and B′ respectively. Then the ratio of area of $\Delta APA'$ and area of $\Delta BPB'$ is
   (A) 1 (B) 2
   (C) 3 (D) 4

Rough work

6. If $\sin x + \sin y \geq \cos\alpha \cos x \ \forall\ x \in R$ then $\sin y + \cos\alpha$ is equal to

(A) $\frac{1}{2}$ (B) 1

(C) 2 (D) –1

7. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}}\,dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}}\,dx$. Then which one of the following is true?

(A) $I > \frac{2}{3}$ and J > 2 (B) $I < \frac{2}{3}$ and J < 2

(C) $I < \frac{2}{3}$ and J > 2 (D) $I > \frac{2}{3}$ and J < 2

8. Tangent to hyperbola $xy = c^2$ at point P intersects the x–axis at T and the y–axis at T′. Normal to hyperbola at P intersects the x–axis at N and the y–axis at N′. If the area of the triangles PNT and PN′T′ are Δ and Δ' respectively then $\frac{1}{\Delta} + \frac{1}{\Delta'}$, is equal to

(A) c^2 (B) $\frac{2}{c^2}$

(C) $\frac{1}{c^2}$ (D) $\frac{c^2}{2}$

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. If $2\sin^2\left(\frac{\pi}{2}\cos^2 x\right) = 1 - \cos(\pi \sin 2x)$, $x \neq (2n+1)\frac{\pi}{2}$, $n \in I$, then

(A) cos 2x is $\frac{3}{5}$ (B) cos 2x is $\frac{1}{2}$

(C) tan x is $\frac{1}{2}$ (D) tan x is $\frac{1}{3}$

Rough work

10. A circle having centre at C is made to pass through the point P(1, 2), touching the straight lines $7x - y = 5$ and $x + y + 13 = 0$ at A and B respectively, then
(A) area of quadrilateral ACBP is 100 sq. units (B) radius of smaller circle is $\sqrt{50}$
(C) area of quadrilateral ACBP is 200 sq. units (D) radius of smaller circle is 10

11. If $ax^2 + bx + c = 0$ has no real root and $a + b + c < 0$ then
(A) $4a - 2b + c > 0$ (B) $4a - 2b + c < 0$
(C) $13a + 5b + 2c < 0$ (D) $5b - 25a - c > 0$

12. Let $f(x) = (x + |x|)\,|x|$, then for all x
(A) f is continuous (B) f′ is differentiable $\forall\ x \in R$
(C) f′ is continuous (D) f″ is continuous

Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 13 to 15

Read the following write up carefully and answer the following questions:
At times the methods of coordinates becomes effective in solving problems of properties of triangles, we may choose one vertex of the triangle and one side passing through this vertex as x–axis. Thus, without loss of generality, we can assume that every triangle ABC has a vertex B situated at B(0, 0), vertex C at (a, 0) and A as (h, k).

13. If in ΔABC, AC = 3, BC = 4 medians AD and BE are perpendicular, then area of triangle ABC must be equal to
(A) $\sqrt{7}$ (B) $\sqrt{11}$
(C) $2\sqrt{2}$ (D) $\sqrt{13}$

Rough work

14. Suppose the bisector AD of the interior angle A of ΔABC divides sides BC into segments BD = 4, DC = 2. Then we must have
(A) b > c and c < 4 (B) b $\in$ (2, 6) and c < 1
(C) b $\in$ (2, 6) and c $\in$ (4, 12) (D) none of these

15. If altitude, CD = 7, AE = 6 and E divides BC given that $\frac{BE}{EC} = \frac{3}{4}$ then c must be
(A) $2\sqrt{3}$ (B) $5\sqrt{3}$
(C) $\sqrt{5}$ (D) $4\sqrt{3}$

Paragraph for Question Nos. 16 to 18

Read the following write up carefully and answer the following questions:

Let $f(x) = 12x^2 \int_0^1 yf(y)\,dy + 20\int_0^1 xy^2 f(y)\,dy + 4x$

16. The maximum value of f(x) is
(A) 8 (B) $\frac{1}{8}$
(C) 16 (D) $\frac{1}{16}$

17. The number of solutions of the equation $|f(|x|)| = e^{|x|}$
(A) 0 (B) 2
(C) 4 (D) 3

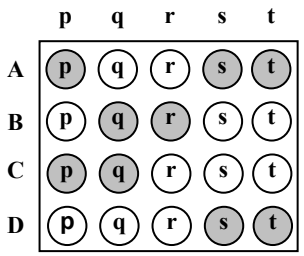
18. The range of $f(-2^x)$ is
(A) $(-\infty, 0)$ (B) $(0, \infty)$
(C) $\left(-\infty, \frac{1}{8}\right)$ (D) $\left(\frac{1}{8}, \infty\right)$

Rough work

SECTION – B

(Matrix Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Let $f(x)=\begin{cases}[x] & ; \ x \in [-2, 0) \\ |x| & ; \ x \in [0, 2]\end{cases}$; where [.] represent G.I.F. and g(x) = sec x, $x \in R - (2n + 1)\frac{\pi}{2}$.

Match the following statements in column–I with their values in column–II in interval $\left(-\frac{3\pi}{2}, \frac{3\pi}{2}\right)$.

	Column – I		Column – II
(A)	Limit of fog exist at	(p)	–1
(B)	Limit of gof does not exist at	(q)	π
(C)	Points of discontinuity of fog is/are	(r)	$\frac{5\pi}{6}$
(D)	Points of differentiability of fog is/are	(s)	$-\pi$
		(t)	$\frac{\pi}{3}$

2. Match the following column–I with column–II.

	Column – I		Column – II
(A)	One ball is drawn from a bag containing 4 balls and is found to be white. The events that the bag contains 1 white, 2 white, 3 white and 4 white balls are equally likely. If the probability that all the balls are white is $\frac{p}{15}$ then the value of p is	(p)	9
(B)	From a set of 12 persons if the number of different selection of a committee, its chairperson and its secretary (possibly same as chairperson) is $13 \cdot 2^{10}$m then m is	(q)	3
(C)	If x, y, z > 0 and x + y + z = 1, then the least value of $\frac{5x}{2-x}+\frac{5y}{2-y}+\frac{5z}{2-z}$ is	(r)	8
(D)	If $\sum_{K=1}^{12} 12K \cdot {}^{12}C_K \cdot {}^{11}C_{K-1}$ is equal to $\frac{12\times 21\times 19\times 17\times .....\times 3}{11!}\times 2^{12}\times p$ the p is	(s)	6
		(t)	12

Rough work